

Yuqian Zhang

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EDUCATION

Duke Kunshan University (DKU) & Duke University Dual Degree	08/2022-05/2026
<i>B.S. in Applied Math & Computational Science; Math Track (DKU)</i>	Kunshan, China
<i>B.S. in Interdisciplinary Studies: Applied Math & Computational Science; Math (Duke)</i>	Durham, U.S.
- GPA: 3.85/4.0; Dean's List Fall 2023, Spring 2024, Spring-Fall 2025 (w. Distinction) Latin Honors – magna cum laude - Courses: Stochastic Differential Equation, Measure Theory, Numerical Analysis, Advance Linear Algebra, Algorithm Analysis & Design, Graphics Software Architecture, Image Data Science.	

RESEARCH EXPERIENCE

<i>Research Intern with Dr. Yi Zhang, Professor of Neurosurgery</i>	12/2024-Now
Lab: Home Yi Zhang CompBio @ Duke yi-zhang-compbio-lab	Durham, U.S.

- PinFuse: Spatial Transcriptomics Data Denoising through GVAE and Mass Conservation**
 - Developed **PinFuse**, a lightweight Variational Graph Auto-Encoder (GVAE) with Position-induced Transformer (PiT) variants for denoising spatial transcriptomics (ST) data. Proposed a neural advection-diffusion operator that preserves localized graph structure and mass conservation while modeling zero-inflated negative binomial observations; incorporated Pyro for probabilistic inference and achieved improved gene reconstruction, latent clustering, and tissue purity over recent methods including ResolVI.
 - Work will be published this year.**
- Gene Expression Field Estimation via Poisson Resampling and Neural ODEs**
 - Formulated ST data as a spatial Poisson sampling process over a diffused gene expression field and developed a reversed Neural ODE framework for field reconstruction from sparse observations, combining supervised pretraining with semi-supervised operator learning.

<i>Research Assistant with Dr. Zuchuan Li, Professor of Data and Environmental Sciences</i>	11/2023-Now
	Kunshan, China

- Learning the Joint Distribution from Missing Values and Flow Matching and Diffusion**
 - Explored recent works of Diffusion Transformer (DiT) and Just-image Transformer (JiT). Derived Diffusion Posterior Sampling (DPS) algorithm on Flow Matching. Implemented MissingJiT and MissingDiT with mask attention to estimate the joint distribution from partially-observed satellite data and batched missing tokens with dynamic length.
 - Work will be published this year.**
- Solving Time Series Inverse Problem by Combining Bayesian-PINN with Kalman Filter**
 - Developed a Bayesian PINN–Kalman Filter framework for time-series inverse problems, including a patched covariance method to improve filtering performance; implemented a CUDA-accelerated 2D advection-diffusion solver as a neural operator in PyTorch.
 - Conducted an independent study of score-based diffusion models for inverse problems, deriving and implementing SDE-based diffusion processes and Diffusion Posterior Sampling (DPS); reproduced DDPM and other sampling methods for both linear and nonlinear inverse problems.

PROFESSIONAL EXPERIENCE

WebGL Implementation of Reflective Gaussian Splatting (Ref-Gaussian)	08/2025-03/2026
<i>Signature Work</i>	Kunshan, China

- Surveyed recent 3D Gaussian Splatting methods and implemented Ref-Gaussian in WebGL, achieving 60+ FPS real-time browser rendering with optimized loading and memory usage.
- Proposed Approximate Tangent Axis (ATA) estimation for Gaussian Surfels, improving position-map smoothness and rendering stability; combined with alpha discard, frustum culling, and lazy updates to improve throughput and reduce GPU usage. Built an end-to-end COLMAP, Ref-Gaussian, post-processing, Docker deployment pipeline for scalable browser-based rendering.
- [XDzzzzzyq/dku-splat: WebGL implementation of Ref-Gaussian for Campus Navigation](#)

Blender Development	02/2024-Now
<i>Individual Contributor; Blender</i>	Remote
- Enhanced Blender's curve sculpting and geometry editing pipeline, implementing curve radius initialization, interpolation, UI integration, and related documentation; also improved several material property editing and curve-mapping interactions. 5 pull requests accepted into the official Blender repository. Blender Contribution Reports - Blender Projects	
CUDA IP Operator Development	9/2024-12/2024
<i>Independent Project</i>	Kunshan, China
- Developed a CUDA-accelerated Python library for Image Processing (IP) and Computer Vision, achieving high-throughput GPU performance and scalable execution. Implemented core algorithms including convolution, geometric transformations, dithering, morphological operations, and texture segmentation. XDzzzzZyq/computer-vision: CS 207 Homeworks	
GPU(OpenGL) based Renderer Development	05/2022-06/2026
<i>Independent Project</i>	China
- Led long-term development of a real-time rendering engine for over 4 years, producing 30,000+ lines of C++/OpenGL code across 600+ commits. Studied 30+ foundational and state-of-the-art graphics papers, including reproductions of 3 SIGGRAPH works. - Conducted in-depth exploration of physically based rendering, high-quality shadowing, signed distance fields, spatio-temporal denoising, and real-time global illumination. XDzzzzZyq/Opengl: OpenGL renderer development by XDzZyq	
TEACHING	
CS101 Teaching Assistant	2025 Fall
- Led weekly office hours and lab sessions to support student learning. Guided students in programming fundamentals, debugging, and problem-solving. - Included 12 hours per month of coaching and instruction for tens of students	
MATH105 Teaching Assistant	2025 Fall
Led weekly office hours and recitation sessions to support student learning, reinforcing mathematical foundations, conceptual understanding, and rigorous problem-solving skills.	
AWARDS & SCHOLARSHIPS	
DKU Student Experiential Learning Fellowship Program	2025 Summer
Awarded funding to conduct a summer research project with Prof. Yi Zhang at Duke, culminating in a written report, poster, and presentation.	
Duke Undergraduate Semester Research Program Reward	2025 Spring
Awarded funding to conduct a semester-long research project with Prof. Yi Zhang at Duke, culminating in a written report.	
DKU Summer Research Scholars	2024 Summer
Awarded funding to conduct a summer research project with Prof. Zuchuan Li at DKU, culminating in a written report, poster, and presentation.	
Mathematical Contest in Modeling 2024 C – Honorable Mention	2024 Spring
Proposed Psychological Angular Momentum Model (PAMM), a machine learning method to evaluate athletes' in-game performance and momentum. Integrated physical dynamics with time series model.	
MathorCup 2023 – Big Data Competition – Second Award	2023 Winter
	Kunshan, China
ADDITIONAL INFORMATION	
Languages: Chinese Mandarin (Native), English (Proficient), Japanese (Beginner)	
Skills: Python (Pytorch), C++, CUDA, OpenGL, Blender, After Effects, Photoshop.	
Interests: Computational Physics/Biology, Machine Learning/Deep Learning, Computer Graphics/Vision.	